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OBD-II connector (Courtesy: M.M. Minderhoud, Wikipedia)

diagnostic trouble codes, data parameters, identification data, input/output parameters and communication parameters.

Data security and functional safety

At one point of time, security in a car meant preventing break-ins and more complex locks. But, security has many more dimensions now.

"With the tremendous amount of data streaming through today's vehicles, security has become a necessity for automakers to protect their control systems from software attacks," says Ganesh.

Apart from physical intrusion, automakers also need to protect the car from inappropriate manipulation such as car tuning and fake spares. They need to prevent hackers from accessing the various ECUs. New payment methods for parking fees or tolls bring up the need to secure the flow of transactional data. Navigation systems pose a threat to privacy. In short, there are umpteen complex dimensions to automotive security today.

Hence a combination of security solutions including chip cards and identification systems for entry control, hardware and software security mod-

ules to prevent unauthorised access to the engine control unit, tamper detection to protect against unauthorised code changes, power modifications and emissions tampering, and advanced network security to protect streaming data is used to protect a vehicle. Crypto and authentication chips, such as those offered by NXP, can bring the credit card or electronic passport level of security to cars.

ISO 26262 standard helps enforce functional safety requirements in today's automobiles. It not only helps design safe products but also set up appropriate processes for developing such products. It is being increasingly adopted by automotive system developers.

Road safety

Features such as airbags and stability control systems, adaptive cruise control, anti-lock braking system, lane-departure warning, tyre-pressure sensor systems, panoramic rear-view aids, traffic light recognition, collision detection, electronic steering and chassis domain control are used in various combinations to improve passenger safety.

Such safety mechanisms involve a complex interplay of intelligent sen-